

A CubeSat (u-class spacecraft) is a type of miniaturised satellite for Space research that is made up of multiples of $10\text{cm} \times 10\text{cm} \times 11.35\text{cm}$ cubic units. It is no bigger than a large toaster, but batteries occupy considerable space in it. Placing a normal battery in an experiment at NASA takes up 20 to 35 per cent of the available volume. With this development, the battery can now be placed with the payload structure, providing more space to the scientists.

This technology could be used on satellite structural trusses, on International Space Station or to power habitat structures established on another planet. Commercial applications could include automobile frames or tabletop battery rechargers. The batteries can be made to be impact-, moisture- and flame-resistant with proper reinforcement, further increasing their scope of utilisation.

Sticky insect-sized drones to pollinate crops

Japanese scientists have developed tiny insect-sized drones coated with horse hair and a sticky gel that may help pollinate crops in future and offset the costly decline of bee populations worldwide.



The bio-inspired flying robotic artificial pollinator (Image courtesy: www.sciencedirect.com)

Undersides of these artificial pollinators are coated with horse hair and an ionic gel that is sticky enough to pick up pollen from one flower and deposit it onto another. The researchers are hopeful that their invention could someday help carry the burden that modern agricultural demand has put on colonies and, in turn, benefit farmers.

"The findings, which will have applications for agriculture and robotics, could lead to the development of artificial pollinators and help counter the problems caused

by declining honeybee populations," says Eijiro Miyako, chemist at National Institute of Advanced Industrial Science and Technology (AIST) Nanomaterial Research Institute in Japan.

To determine whether the gel could grasp onto pollen, Miyako collected ants, put the ionic goop droplet on their bodies and left them to roam free in a box of tulips. Ants with the test material on their back gathered much more pollen than those without gel.

In separate experiments using houseflies, the gel was also found to have a camouflage effect—changing colour in response to different sources of light—which could help artificial pollinators avoid predation.

Smart needle that makes brain surgery safer

University of Adelaide scientists have developed a smart needle with a tiny camera that allows surgeons to view at-risk blood vessels and avoid injuring these—an advance that may make brain surgery safer. The tiny imaging probe, encased within a brain biopsy needle, allows surgeons to see blood vessels as they insert the needle, allowing them to avoid causing bleeds that can potentially be fatal.

Robert McLaughlin, professor at University of Adelaide, says, "We call it a smart needle. It contains a tiny fibre-optic camera, the size of a human hair, shining infrared light to see the vessels before the needle can damage these." "And what is really exciting is the computer smarts behind this so that the computer itself recognises the blood vessel and alerts the surgeon," he adds.

"To have a tool that can see blood vessels as we proceed through the brain would revolutionise neurosurgery. It will open the way for safer surgery, allowing us to do things we have not been able to do before," said Christopher Lind, consultant neurosurgeon at Sir Charles Gairdner Hospital and University of Western Australia.

Over the past six months, the smart needle has been used in a pilot trial with 12 patients undergoing neurosurgery at Sir Charles Gairdner Hospital in Western Australia.

Ingestible devices that draw power from stomach acid

Researchers from Brigham and Women's Hospital and Massachusetts Institute of Technology have developed a set of ingestible devices that use stomach acid for long-term power. They have demonstrated a small voltaic cell that is sustained by the acidic fluids in the stomach. The system